

Star Spots of AU Mic with Starry

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Star Spots

Starspots are features on stars that are analogous to sunspots on the sun. Starspots are an area on the surface of a star that is darker than its surroundings. Star spots are areas of strong magnetic flux, likely generated from the an star's convection zone. They are important to study because starspots can give us a look into the

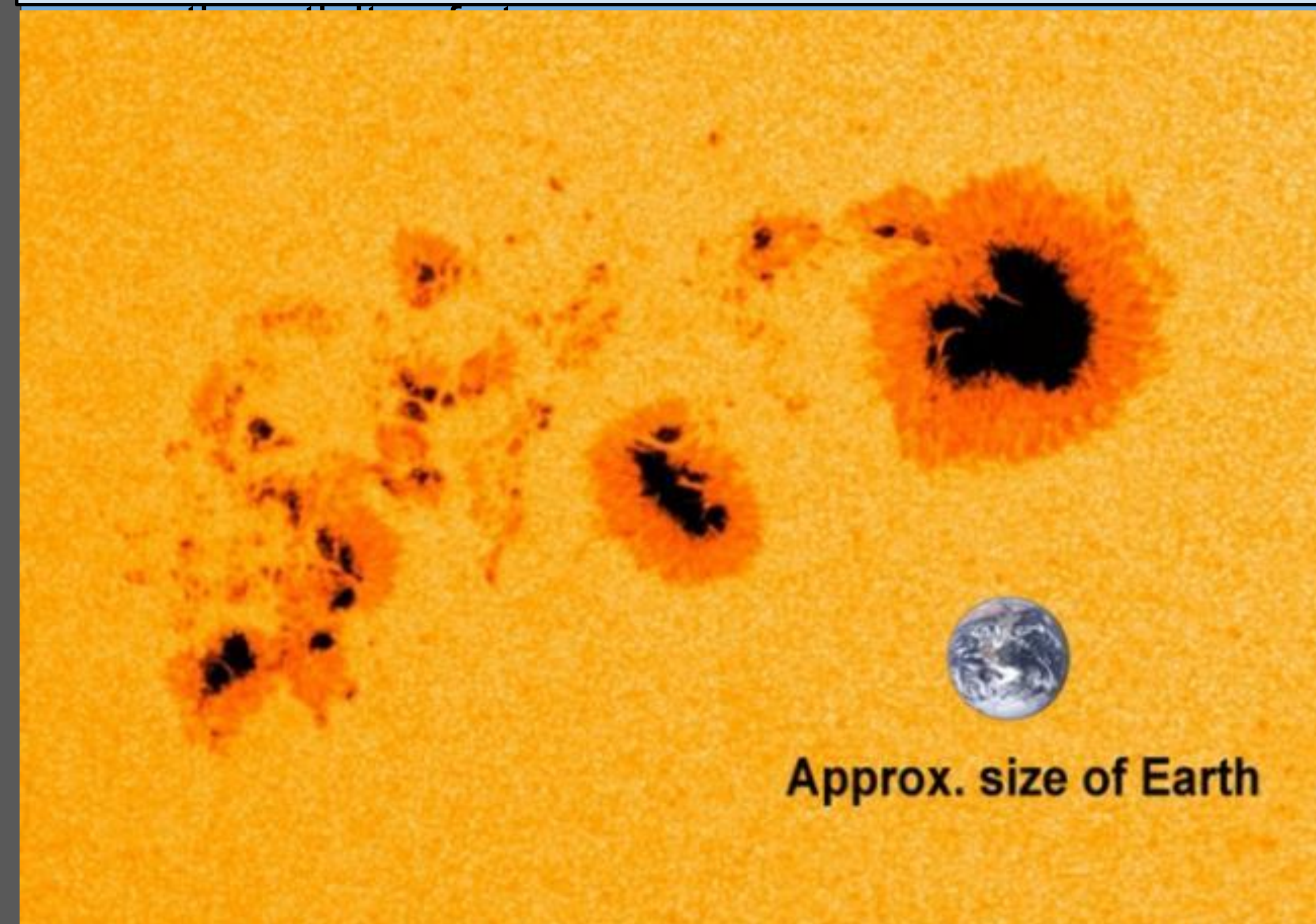


Fig 1: One of the largest sunspots with the Earth as a comparison.

Spot Filling Fraction

Previous work has been done on constraining the star-spot filling fractions on 240 stars in the Pleiades and M67 star clusters (Cao et al. (2022)). This was done using APOGEE spectra. A spectroscopic pipeline was developed which solves for star-spot filling fractions and star-spot temperature contrast. A two temperature model was used. The data for these measurements came from APOGEE and Gaia DR3. Star-spots properties are also presented for AU Mic (Waalkes et al. 2024) by combining photometry and spectra. AU Microscopii (AU Mic) is an active M dwarf which has had its starspot data studied.

AU Mic

AU Mic is an active 24 Myr old pre-main-sequence M dwarf with a rotation period of 4.86 days. Its starspots were studied in order to facilitate current and future studies of its planets, AU Mic b and c. The Las Cumbres Observatory was used to gather this data. With the advent of JWST, we are able to study exoplanet atmospheres in more detail. These types of studies will help to sharpen our tools in order to be able to study these atmospheres much better. Waalkes et al. (2024) presents an artist representation of spots below.

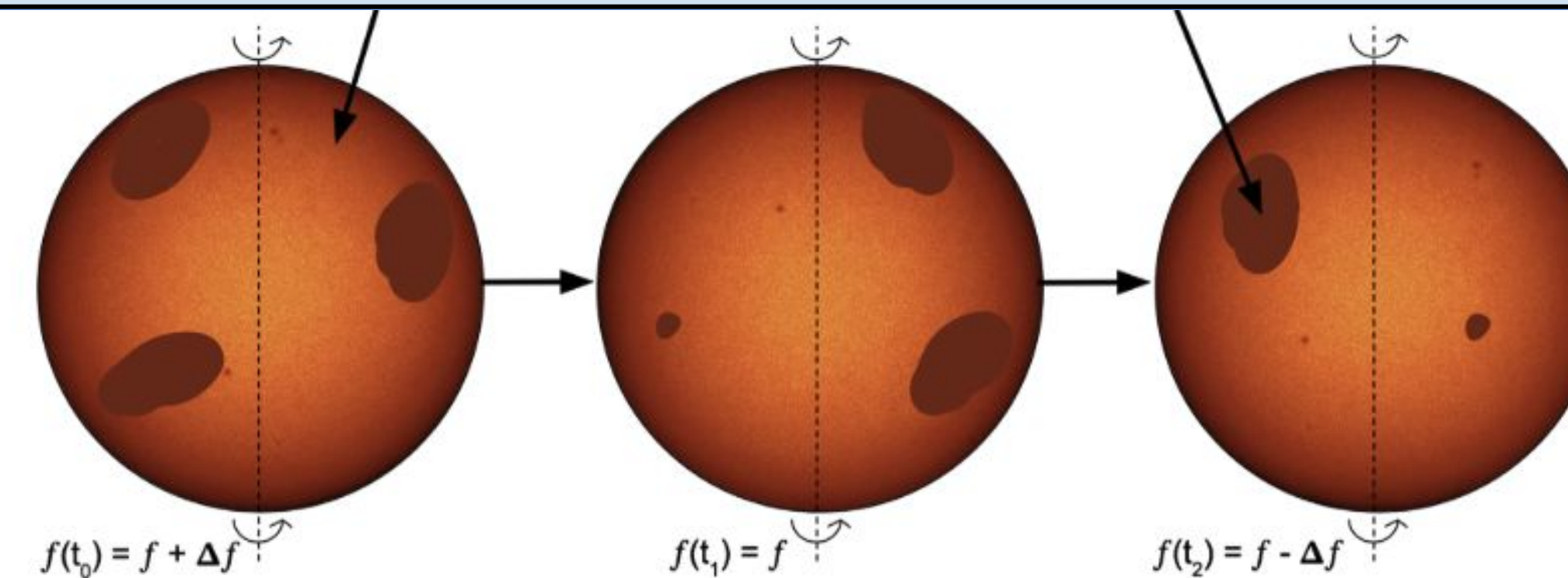


Fig 2: AU Mic and an artist representation of its starspots.

Starry

The aim of this project will be to use light curve mapping techniques on existing light curves of stars to reveal the structure of starspots. This will allow us to maximize the use of light curves. We use Starry, a code developed by Luger et al. (2019), which has primarily been used on stellar light curves. Starry is a python package which uses light curves as an input to generate surface maps. Starry is a powerful tool in that it allows us to use already existing data in order to study light curves.

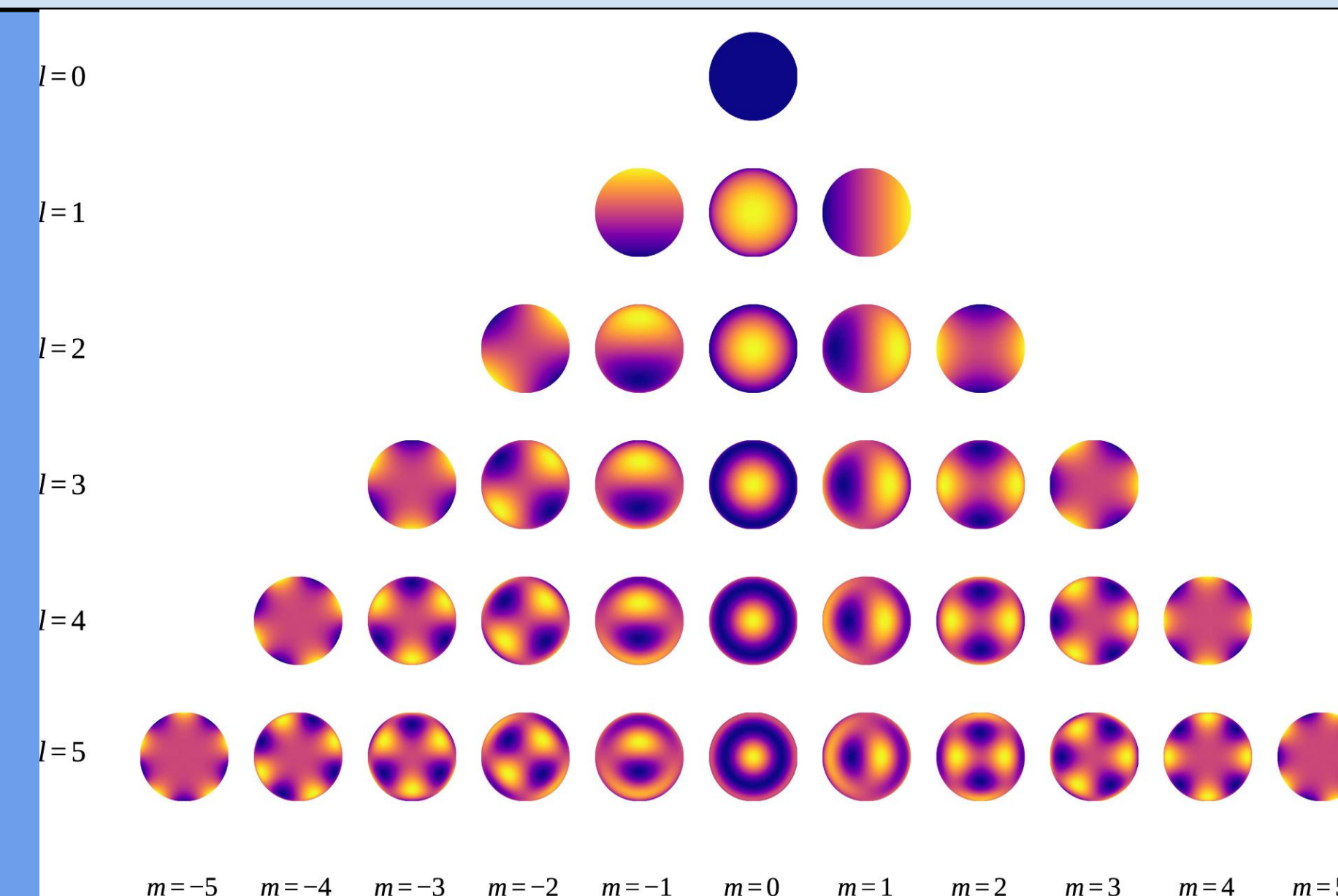


Fig 3: Starry uses spherical harmonics to generate surface maps.

Results

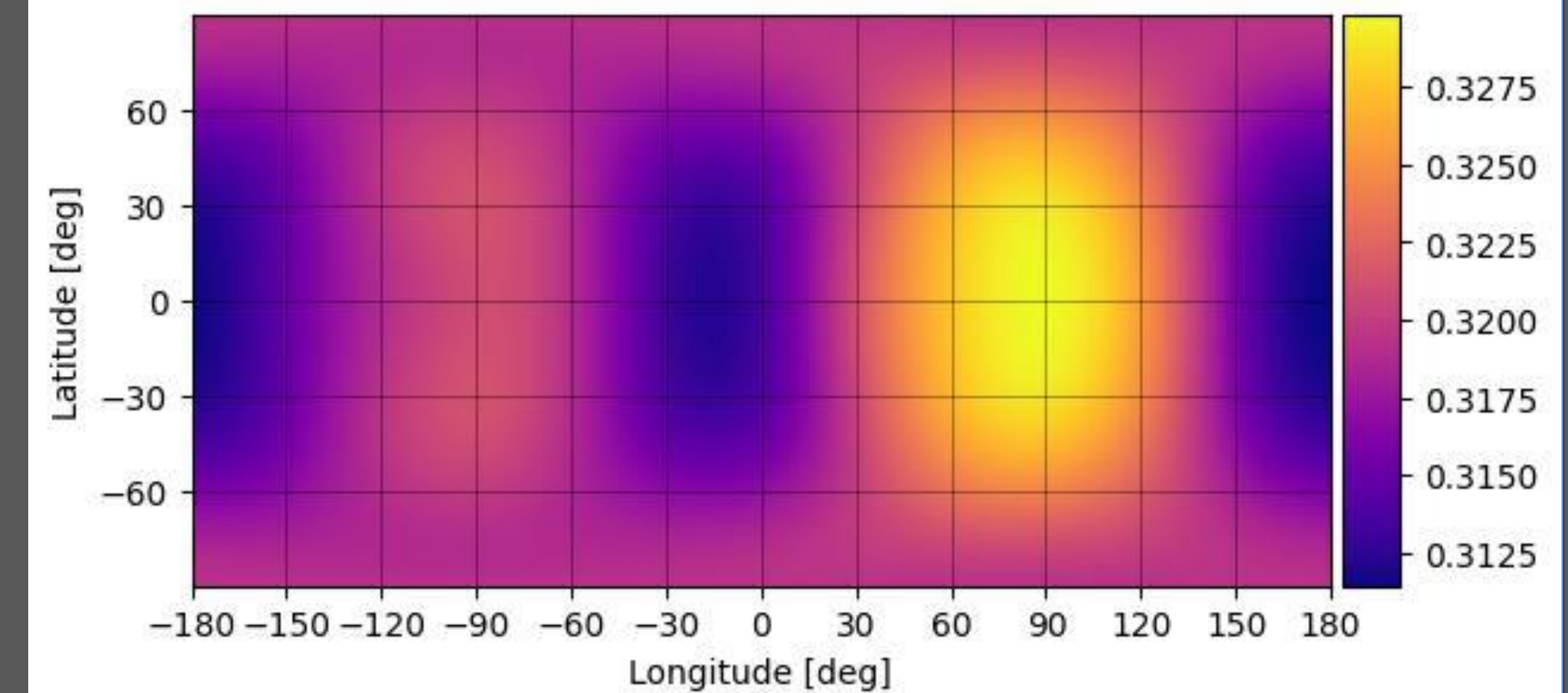


Fig 4: AU Mic's surface as calculated by Starry

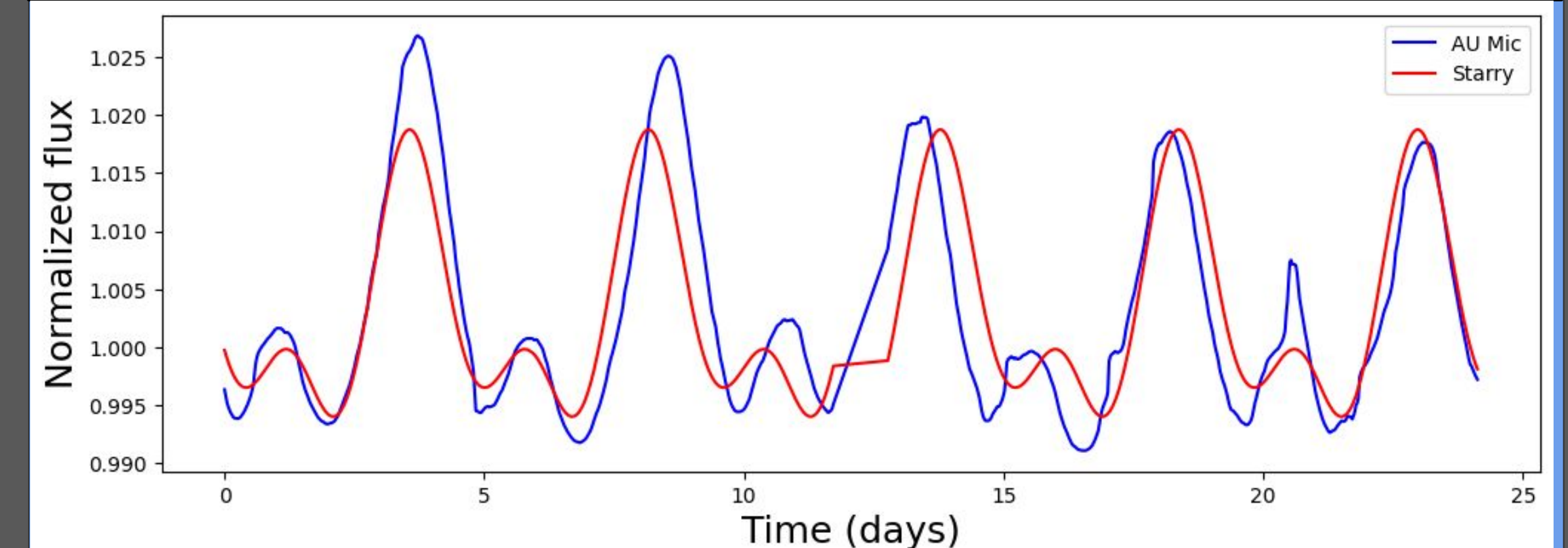


Fig 5: Binned AU Mic light curve in TESS and simulated Starry light curve.

The light curves generated by starry show a discrepancy with the flux compared to the original observed light curve. Also, compared to the fleck results, Starry shows spots with do not vary latitudinally. Fleck also favors polar spots, while Starry does not.

Conclusion

Starry provides us with another method to generate star spots, which we can compare with fleck's. Next steps will be to analyze each individual rotation of the light curve in Starry.

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